

Notes: Dix-Carneiro and Kovak (AER, 2017)

Trade Liberalization and Regional Dynamics

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1 Big picture

- **Question.** How do local labor markets adjust over time after a major trade liberalization? The paper asks whether trade shocks have only short- and medium-run regional effects, or whether their effects keep evolving for many years.
- **Setting.** The paper studies Brazil’s early-1990s trade liberalization. Tariffs fell sharply and unevenly across industries, and regions differed in their pre-liberalization industry mix.
- **Main empirical object.** The authors construct a *regional tariff reduction* measure, RTR_r , which is larger for regions whose initially important industries experienced larger tariff cuts.
- **Core data advantage.** The main data are the RAIS administrative employer-employee records from 1986 to 2010. These data allow the authors to observe formal employment and earnings every year for 25 years and to trace annual adjustment paths after the shock.
- **Main result.** Regions facing larger tariff reductions suffered persistent relative declines in formal-sector earnings and employment. The earnings effect after 20 years is more than three times the effect after 10 years.
- **Why this is surprising.** Standard spatial equilibrium logic predicts that migration and labor reallocation should gradually arbitrage away regional earnings differences. Instead, the regional earnings gaps grow for more than a decade.
- **Mechanism.** The paper finds little evidence that population migration, worker composition, or slow import/export responses explain the growing effects. The supported mechanism is imperfect interregional labor mobility combined with dynamic labor demand, driven by slow capital adjustment and agglomeration economies.
- **Economic takeaway.** Short-run estimates of local trade-shock effects can seriously understate long-run regional consequences. Trade liberalization’s local labor-market costs remain unevenly distributed across space even two decades after the policy change.

2 Incremental contribution of the paper

- **From static local effects to dynamic adjustment.** Prior local-labor-market trade papers often measure effects over a single window, usually around seven to ten years. This paper traces the year-by-year impulse response for nearly two decades after liberalization.
- **A clean policy shock.** Brazilian liberalization was a discrete and largely completed policy change from 1990 to 1995. This makes it easier to separate adjustment dynamics from the arrival of new shocks.
- **Administrative data contribution.** RAIS allows annual, fine-geographic measurement of formal employment, earnings, establishments, job creation, job destruction, entry, and exit.
- **A challenge to spatial arbitrage intuition.** The paper shows that regional earnings effects grow rather than shrink. This is the opposite of the usual prediction that labor mobility should reduce local wage gaps over time.

- **Mechanism contribution.** The paper separates many possible explanations: urban decline, changing worker composition, slow trade-flow responses, post-liberalization shocks, capital adjustment, and agglomeration.
- **Dynamic labor demand result.** The paper argues that labor demand itself shifts gradually after trade liberalization because firms and capital do not instantly leave adversely affected regions, and because local economic activity can feed back into productivity.
- **Quantitative contribution.** The paper does not only document the mechanism qualitatively. It uses a model-based decomposition to show that slow capital adjustment and plausible agglomeration elasticities can account for the long-run magnitude of the earnings effects.
- **Policy contribution.** The results suggest that trade-adjustment assistance and regional policies need to account for very long adjustment horizons, not just temporary dislocation immediately after trade reform.

3 Data and measurement

3.1 Brazilian trade liberalization

- Brazil had a long history of high trade barriers under import-substitution policies. Liberalization effectively began in 1990 and was largely complete by 1995.
- Average tariffs fell substantially, from roughly 30.5% to 12.8%, and remained relatively stable afterward.
- The tariff cuts varied strongly by industry. Some sectors, such as agriculture and mining, experienced small changes, while sectors such as apparel and rubber had much larger cuts.
- The authors measure industry liberalization using the long difference in $\ln(1 + \tau_i)$ from 1990 to 1995, where τ_i is the industry tariff.

Interpretation: The key identifying shock is not that Brazil as a whole liberalized, but that different industries were exposed to different tariff reductions, and different regions specialized in different industries before the reform.

3.2 Regional labor markets

- The main local labor market unit is the Brazilian microregion, slightly aggregated where needed to maintain consistent boundaries over time.
- The main RAIS-based analysis uses 475 local labor markets. Census-based analyses with earlier data and informal-sector outcomes use 405 markets.
- Microregions group economically integrated contiguous municipalities and are meant to approximate local labor markets.
- The paper includes state fixed effects in the main specifications, so the identifying variation is largely within-state differences in exposure to tariff cuts.

Interpretation: The design compares regions in the same broad institutional environment, but with different pre-liberalization industry composition.

3.3 Administrative and Census data

- RAIS is an administrative census of Brazil’s formal labor market, covering workers with signed work cards.
- RAIS includes worker identifiers, establishment identifiers, municipality, industry, sex, age, education, and earnings. This allows worker and establishment tracking over time.
- RAIS does not observe informal workers, self-employed workers, the unemployed, or people out of the labor force.
- The paper therefore supplements RAIS with the Brazilian Demographic Census, which covers the whole population and allows analysis of informal employment, informal earnings, and population.

Interpretation: The RAIS data make the dynamic formal-sector analysis possible; the Census data are essential for checking whether formal-sector declines reflect migration or shifts into informality.

3.4 Regional tariff reduction measure

The regional shock is a weighted average of industry tariff reductions:

$$RTR_r = - \sum_i \beta_{ri} d \ln(1 + \tau_i), \quad (1)$$

where $d \ln(1 + \tau_i)$ is the 1990–1995 change in the industry tariff term. The minus sign means that a larger value of RTR_r corresponds to a larger tariff reduction.

The weights follow the specific-factors logic:

$$\beta_{ri} = \frac{\lambda_{ri}/\varphi_i}{\sum_j \lambda_{rj}/\varphi_j}, \quad (2)$$

where λ_{ri} is the initial share of regional labor in industry i , and φ_i is the cost share of nonlabor factors.

Interpretation: A region is more exposed if its important pre-liberalization industries had larger tariff cuts and if the model implies that industry shocks transmit strongly to local labor demand.

3.5 Outcome variables

- The main earnings outcome is a regional formal earnings premium, estimated after controlling for worker age, sex, education, and industry.
- The main employment outcome is regional log formal employment.
- Additional outcomes include working-age population, informal employment, informal earnings, number of formal establishments, average establishment size, establishment entry and exit, and job creation and destruction.

- The paper also constructs a real earnings measure using local housing rents to infer local price changes, following a regional deflator approach.

Interpretation: The outcome set allows the paper to distinguish wage effects, employment effects, population mobility, movement into informality, firm dynamics, and capital adjustment.

4 Models and empirical specifications

4.1 Regional labor demand shock

The paper builds on a specific-factors model of regional economies. In this framework, industry price changes caused by tariff reductions shift regional labor demand. A region’s exposure depends on the industries present before liberalization. The theoretical shock is a weighted average of industry price changes:

$$\hat{w}_r = \sum_i \beta_{ri} \hat{P}_i, \quad (3)$$

where $\hat{P}_i$ is the proportional price change facing industry i . The empirical RTR_r replaces the unobserved price change with the tariff change.

Interpretation: The model predicts that tariff cuts reduce labor demand more in regions specialized in industries whose output prices fall more.

4.2 Annual regional response regression

The central empirical specification is estimated separately for each post-liberalization year t :

$$y_{rt} - y_{r,1991} = \theta_t RTR_r + \alpha_{st} + \gamma_t(y_{r,1990} - y_{r,1986}) + \varepsilon_{rt}. \quad (4)$$

- y_{rt} is a regional outcome, such as a regional earnings premium or log formal employment.
- RTR_r is fixed over time and measures the 1990–1995 regional tariff reduction.
- α_{st} are state fixed effects, allowed to differ by year.
- The pretrend control accounts for pre-liberalization changes in the outcome.
- The sequence θ_t traces the cumulative effect of liberalization by year t .

Interpretation: The coefficient path θ_t is an empirical impulse response. If spatial arbitrage eliminates local effects, θ_t should shrink in magnitude over time. Instead, the paper finds that earnings effects grow.

4.3 Regional earnings premia

Regional earnings premia are estimated in each year with a worker-level regression:

$$\ln(earn_{jrit}) = X_j\Gamma_t + \phi_{it} + \mu_{rt} + \epsilon_{jrit}. \quad (5)$$

- X_j includes flexible controls for age, sex, and education.
- ϕ_{it} are industry fixed effects.
- μ_{rt} are the regional earnings premia used as outcomes.

Interpretation: The paper is not simply comparing raw average wages. It compares regional earnings after accounting for observable worker composition and industry mix.

4.4 Worker-composition checks

To test whether changing unobserved worker composition explains the results, the paper estimates earnings premia with worker fixed effects:

$$\ln(earn_{jairt}) = \alpha_j + \psi_a + \phi_{it} + \mu_{rt} + \epsilon_{jairt}. \quad (6)$$

It also estimates a more flexible specification in which returns to worker characteristics can vary over time:

$$\ln(earn_{jairt}) = \delta_t\alpha_j + \psi_a + \phi_{it} + \mu_{rt} + \epsilon_{jairt}. \quad (7)$$

Interpretation: If the main earnings pattern were caused by high-ability workers selectively leaving formal employment in harder-hit regions, the effect should weaken after controlling for individual fixed effects. It does not.

4.5 Dynamic labor demand mechanism

The paper extends the regional specific-factors model to allow capital and productivity to adjust over time. Production uses labor, capital, and fixed factors, and productivity may depend on local formal economic activity through agglomeration.

A simplified equilibrium relation is:

$$\hat{w}_r = \sum_i \beta_{ri} \hat{P}_i + \text{labor adjustment} + \text{capital adjustment} + \text{agglomeration adjustment}. \quad (8)$$

The key conceptual distinction is:

- **Labor supply adjustment alone** should reduce wage gaps over time.
- **Capital adjustment** can deepen local wage declines if capital slowly leaves harder-hit regions.
- **Agglomeration** can amplify declines because lower formal employment reduces local productivity, which further lowers wages and employment.

Interpretation: The paper's central mechanism is not just that labor moves slowly. It is that labor moves imperfectly while labor demand keeps shifting downward in exposed regions.

5 Identification and empirical design

5.1 Source of identifying variation

- Industry tariff cuts are highly correlated with pre-liberalization tariff levels, which were largely inherited from an older protection structure rather than chosen in response to 1990s regional shocks.
- Regions differ in exposure because they had different industry mixes before liberalization.
- The main specifications include state fixed effects and pre-liberalization outcome trends.
- The paper therefore compares within-state regions with different exposure to a national policy reform.

Interpretation: The identifying assumption is that, conditional on state fixed effects and pre-trends, regions with larger $RT\bar{R}_r$ would not have had systematically different post-1991 outcome paths absent liberalization.

5.2 Why annual dynamics matter

- A single post-period estimate cannot tell whether the shock effect is fading, stable, or growing.
- The RAIS data let the authors estimate the regional effect separately for every year from 1992 to 2010.
- Because the policy shock was largely complete by 1995, later changes in θ_t are interpreted as adjustment dynamics rather than newly arriving tariff shocks.

Interpretation: The paper’s strongest contribution is the time path: the long-run effect is much larger than the medium-run effect.

5.3 Main threats and responses

- **Preexisting regional trends.** The paper controls for 1986–1990 trends and shows robustness to longer Census-based pretrends.
- **Changing worker composition.** The paper uses worker fixed effects and time-varying returns to worker characteristics.
- **Migration.** Census population regressions show no meaningful working-age population response to tariff exposure.
- **Informal-sector shifts.** The paper documents increases in informal employment but no comparable decline in informal earnings.
- **Post-liberalization shocks.** The paper checks later tariffs, exchange rates, privatization, commodity prices, and China-related trade shocks.
- **Slow trade-flow response.** Import and export quantities do not have the timing needed to explain growing earnings effects, and controls/instruments for trade quantities leave the main pattern intact.

Interpretation: The identification strategy is strongest because it combines a clear policy shock with a long panel and a detailed mechanism analysis.

6 Tables and figures

6.1 Figure 1: Tariff Changes

Read:

- Figure 1 plots industry-level changes in $\ln(1 + \text{tariff})$ from 1990 to 1995.
- Most industries have negative bars, indicating tariff cuts.
- The size of the cuts differs greatly by industry.
- Agriculture and mining are close to zero, while apparel, rubber, and several manufacturing sectors face much larger declines.

Economic message: The figure shows the first source of variation: the national trade reform was not uniform across industries. This cross-industry variation is what becomes local labor-market variation once combined with regional industry composition.

6.2 Figure 2: Regional Tariff Reductions

Read:

- Figure 2 maps the regional tariff reduction measure $RT\bar{R}_r$.
- Lighter regions are more exposed to tariff reductions.
- The table below the map shows substantial dispersion: the mean is about 0.044, the 10th percentile is 0.002, and the 90th percentile is 0.107.
- The map also shows meaningful variation within states, which is important because the regressions include state fixed effects.

Economic message: Trade liberalization creates a geography of exposure. Regions with different initial industry structures face very different local labor demand shocks even though the policy is national.

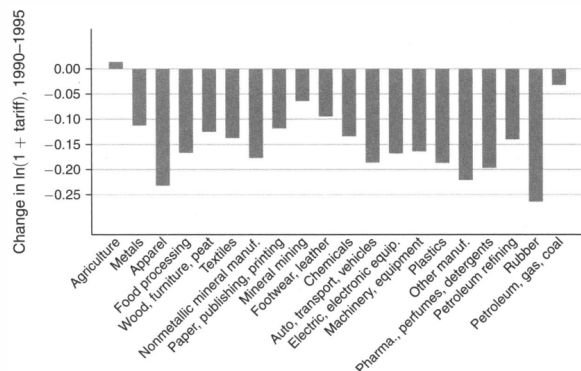


FIGURE 1. TARIFF CHANGES

(a) Figure 1: tariff cuts by industry

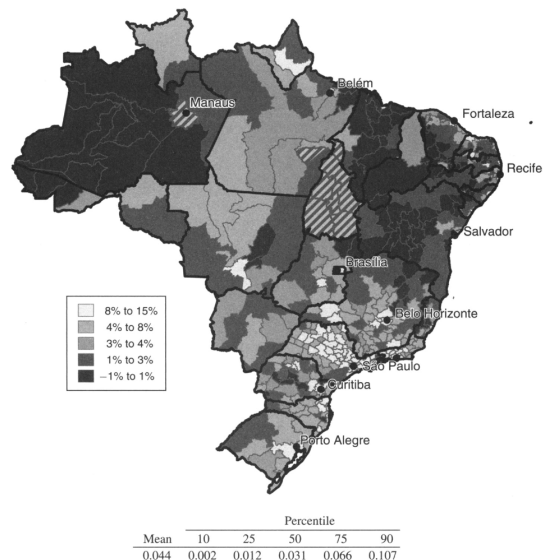


FIGURE 2. REGIONAL TARIFF REDUCTIONS

Notes: Local labor markets reflect microregions defined by IBGE, aggregated slightly to account for border changes between 1986 and 2010. Regions are colored based on the regional tariff reduction measure, RTR_t , defined in (2). Regions facing larger tariff reductions are presented as lighter, while regions facing smaller cuts are shown as darker.

(b) Figure 2: regional tariff reductions

6.3 Table 1: Descriptive Statistics

Read:

- Table 1 summarizes the main shock and outcomes.
- The mean regional tariff reduction is 0.044 with substantial cross-region variation.
- Formal earnings, formal employment, and the number of formal establishments all rise on average over the sample, but the paper studies relative growth across regions.
- Informal and population variables appear only for Census years, because they are not observed in RAIS.

Economic message: The economy as a whole grew, but the empirical question is whether regions exposed to larger tariff cuts grew less than other regions. The table establishes the scale of the variables used throughout the paper.

6.4 Table 2: Regional Formal Earnings and Employment

Read:

- Panel A shows that a larger regional tariff reduction predicts a larger decline in formal earnings premia.
- The 2000 coefficient with state fixed effects and pretrends is -0.529 . For a 10 percentage point larger tariff reduction, this means roughly 5.3 percentage points lower earnings growth by 2000.

- The 2010 coefficient is -1.594 , implying roughly 15.9 percentage points lower earnings growth for the same exposure difference.
- Panel B shows that even after adjusting for local price changes, real earnings effects remain large.
- Panel C shows large formal employment declines, with the coefficient moving from about -3.533 in 2000 to -4.663 in 2010.

Economic message: Table 2 is the central result. The earnings effect grows substantially between 2000 and 2010, contradicting the idea that local wage gaps should fade as workers arbitrage across regions.

TABLE 1—DESCRIPTIVE STATISTICS

	1991–1995	1991–2000	1991–2005	1991–2010	
<i>Panel A. Liberalization shock</i>					
Regional tariff reductions (<i>RTR_t</i>)	0.044 (0.039)				
<i>Panel B. Main outcome variables</i>					
Change in log formal earnings premium	0.258 (0.161)	0.305 (0.174)	0.401 (0.189)	0.712 (0.201)	
Change in log informal earnings premium ^a		−0.050 (0.135)		0.161 (0.197)	
Change in log formal employment	0.268 (0.377)	0.599 (0.549)	0.976 (0.576)	1.308 (0.614)	
Change in log informal employment ^a		0.269 (0.162)		0.291 (0.228)	
Change in log num. formal establishments	0.358 (0.230)	0.728 (0.318)	1.055 (0.389)	1.271 (0.444)	
Change in log avg. formal establishment size	−0.180 (0.279)	−0.260 (0.387)	−0.220 (0.375)	−0.128 (0.384)	
Change in log formal job destruction	−1.014 (0.398)	−0.824 (0.421)	−0.966 (0.466)	−1.135 (0.516)	
Change in log formal job creation	−0.608 (0.387)	−0.099 (0.270)	0.143 (0.221)	0.299 (0.176)	
Change in log formal establishment exit	−1.206 (0.226)	−1.092 (0.285)	−1.183 (0.343)	−1.305 (0.405)	
Change in log formal establishment entry	−0.397 (0.287)	0.052 (0.175)	0.251 (0.139)	0.366 (0.114)	
Change in log working-age population ^a		0.198 (0.103)		0.388 (0.178)	
<i>Panel C. Region characteristics</i>					
	1991	1995	2000	2005	2010
Average formal earnings (2010 R\$)	755.98 (273.08)	1,105.83 (394.71)	944.18 (323.97)	939.93 (480.00)	1,152.40 (469.95)
Formal employment	30,466 (152,267)	34,929 (161,657)	40,100 (163,917)	51,631 (197,206)	70,170 (269,602)
Share agriculture/mining ^a	0.399 (0.194)				
Share manufacturing ^a	0.113 (0.077)				

Notes: 475 microregion observations. See the text for descriptions of all measures.

^aCalculated using the Census, so estimates are available only for 1991, 2000, and 2010.

(a) Table 1: descriptive statistics

TABLE 2—REGIONAL LOG FORMAL EARNINGS PREMIA AND EMPLOYMENT, 2000 AND 2010

Change in outcome:	1991–2000			1991–2010		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. log formal earnings premia</i>						
Regional tariff reduction (RTR_t)	−0.451 (0.152)	−0.638 (0.154)	−0.529 (0.141)	−1.885 (0.316)	−1.736 (0.184)	−1.594 (0.169)
Formal earnings pretrend, 1986–1990			−0.312 (0.149)			−0.418 (0.144)
State fixed effects (26)		✓	✓		✓	✓
R^2	0.040	0.225	0.268	0.320	0.501	0.537
<i>Panel B. log formal real earnings premia: regional deflators following Moretti (2013)</i>						
Regional tariff reduction (RTR_t)				−1.594 (0.306)	−1.382 (0.180)	−1.260 (0.168)
Formal earnings pretrend, 1986–1990						−0.359 (0.133)
State fixed effects (26)					✓	✓
R^2				0.238	0.449	0.477
<i>Panel C. log formal employment</i>						
Regional tariff reduction (RTR_t)	−3.748 (0.516)	−3.545 (0.563)	−3.533 (0.582)	−6.059 (0.560)	−4.675 (0.660)	−4.663 (0.679)
Formal employment pretrend, 1986–1990			−0.033 (0.147)			−0.032 (0.156)
State fixed effects (26)		✓	✓		✓	✓
R^2	0.072	0.291	0.291	0.149	0.409	0.410

Notes: Negative coefficient estimates for the regional tariff reduction imply larger declines in formal earnings or employment in regions facing larger tariff reductions. Microregion observations: panels A and C, 475; panel B, 456 (omits a few sparsely populated locations with insufficient data to calculate regional price deflators). Regional earnings premia calculated controlling for age, sex, education, and industry of employment. Panels A and B: efficiency weighted by the inverse of the squared standard error of the estimated change in log formal earnings premium. Pretrends computed for 1986–1990. Standard errors (in parentheses) adjusted for 112 mesoregion clusters.

(b) Table 2: formal earnings and employment

6.5 Figure 3: Regional Formal Earnings Premia

Read:

- Figure 3 plots annual coefficients on RTR_t for formal earnings premia.
- The pre-liberalization coefficients do not show the same negative pattern; if anything, the post-liberalization decline reverses earlier trends.
- The post-liberalization effect becomes increasingly negative for more than a decade.
- The effect levels off only in the late 2000s.

Economic message: The figure is the clearest visual evidence that trade liberalization's local earnings effects are dynamic and cumulative. The long-run effect is much larger than the medium-run effect.

6.6 Figure 4: Regional Formal Employment

Read:

- Figure 4 plots the corresponding annual coefficients for formal employment.
- Employment effects become sharply negative after liberalization.
- Most of the formal employment divergence is complete by the early-to-mid 2000s.
- The employment profile is consistent with workers leaving formal jobs in harder-hit regions.

Economic message: The employment decline is large and persistent. Together with Figure 3, it implies that both quantities and earnings deteriorate in exposed regions rather than wages recovering through labor reallocation.

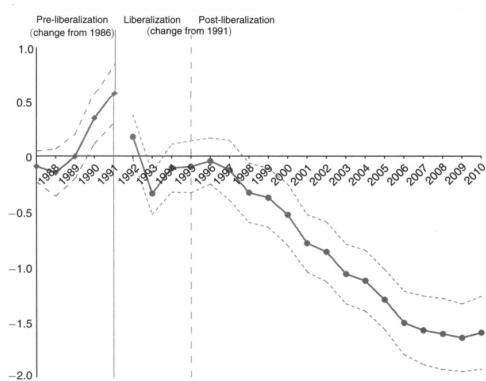


FIGURE 3. REGIONAL LOG FORMAL EARNINGS PREMIA, 1987–2010

Notes: Each point reflects an individual regression coefficient, $\hat{\theta}_t$, following (3), where the dependent variable is the change in regional log formal earnings premium and the independent variable is the regional tariff reduction (RTR_t), defined in (2). Note that RTR_t always reflects tariff reductions from 1990 to 1995. For circles, the earnings changes are from 1991 to the year listed on the x-axis. For diamonds, the changes are from 1986 to the year listed. All regressions include state fixed effects, and post-liberalization regressions control for the 1986–1990 outcome pretrend. Negative estimates imply larger earnings declines in regions facing larger tariff reductions. Vertical bars

(a) Figure 3: formal earnings premia

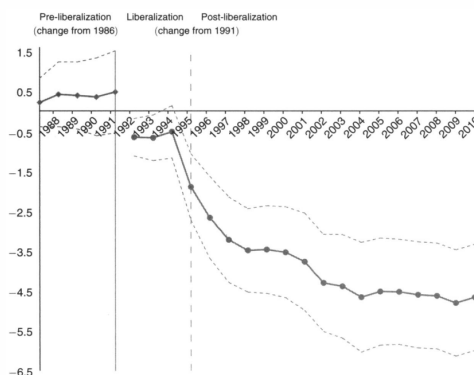


FIGURE 4. REGIONAL LOG FORMAL EMPLOYMENT, 1987–2010

Notes: Each point reflects an individual regression coefficient, $\hat{\theta}_t$, following (3), where the dependent variable is the change in regional log formal employment and the independent variable is the regional tariff reduction (RTR_t), defined in (2). Note that RTR_t always reflects tariff reductions from 1990 to 1995. For circles, the earnings changes are from 1991 to the year listed on the x-axis. For diamonds, the changes are from 1986 to the year listed. All regressions include state fixed effects, and post-liberalization regressions control for the 1986–1990 outcome pretrend. Negative estimates imply larger earnings declines in regions facing larger tariff reductions. Vertical bars

(b) Figure 4: formal employment

6.7 Table 3: Working-Age Population

Read:

- Table 3 uses Census data to test whether working-age population responds to regional tariff reductions.
- Across 2000 and 2010 specifications, the coefficients are small and statistically insignificant after controls.
- The 2010 coefficient with both long pretrend controls is -0.059 , close to zero.

Economic message: Formal employment declines are not primarily due to large out-migration from harder-hit regions. Workers tend to remain in the same region, but many leave formal employment.

6.8 Table 4: Informal Employment and Earnings

Read:

- Panel A shows that informal employment rises in regions facing larger tariff reductions.
- The 2010 coefficient with full controls is 1.196, implying roughly 12 percentage points higher informal-employment growth for a 10 percentage point larger tariff reduction.
- Panel B shows no statistically significant negative effect on informal earnings premia.
- The informal labor market absorbs some workers leaving the formal sector, but it does not show the same falling earnings pattern.

Economic message: The formal employment decline is not simply migration. It partly reflects workers moving into informal employment within the same region. This is central to the paper's interpretation of imperfect interregional mobility.

TABLE 3—REGIONAL LOG WORKING-AGE POPULATION, 2000 AND 2010

Change in log working-age population:	1991–2000			1991–2010		
	(1)	(2)	(3)	(4)	(5)	(6)
Regional tariff reduction (RTR_t)	0.333 (0.243)	−0.061 (0.330)	0.018 (0.204)	0.392 (0.319)	−0.175 (0.473)	−0.059 (0.294)
Population pretrend, 1980–1991	0.406 (0.164)		0.328 (0.171)	0.632 (0.225)		0.531 (0.235)
Population pretrend, 1970–1980		0.297 (0.072)	0.137 (0.047)		0.445 (0.087)	0.190 (0.073)
State fixed effects (26)	✓	✓	✓	✓	✓	✓
R^2	0.654	0.557	0.678	0.666	0.554	0.685

Notes: Positive (negative) coefficient estimates for the regional tariff reduction imply larger increases (decreases) in population in regions facing larger tariff reductions. Outcomes calculated using Census data. There are 405 microregion observations. Efficiency weighted by the inverse of the squared standard error of the dependent variable estimate. Pretrends computed for 1980–1991 and 1970–1980. Standard errors (in parentheses) adjusted for 90 mesoregion clusters.

(a) Table 3: working-age population

TABLE 4—REGIONAL LOG INFORMAL EMPLOYMENT AND EARNINGS PREMIA, 2000 AND 2010

Change in outcome:	1991–2000			1991–2010		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. log informal employment</i>						
Regional tariff reduction (RTR_t)	2.017 (0.431)	1.706 (0.344)	1.593 (0.532)	2.122 (0.468)	1.448 (0.491)	1.196 (0.705)
Informal employment pretrend (1980–1991)	0.069 (0.115)		0.050 (0.114)	0.149 (0.132)		0.109 (0.126)
All employment pretrend, 1970–1980		0.121 (0.056)	0.110 (0.044)		0.263 (0.080)	0.239 (0.063)
State fixed effects (26)	✓	✓	✓	✓	✓	✓
R^2	0.579	0.589	0.592	0.524	0.552	0.562
<i>Panel B. log informal earnings premia</i>						
Regional tariff reduction (RTR_t)	−0.027 (0.161)	−0.217 (0.160)	−0.034 (0.163)	0.352 (0.256)	0.054 (0.298)	0.338 (0.251)
Informal earnings pretrend, 1980–1991	−0.191 (0.049)		−0.193 (0.048)	−0.288 (0.086)		−0.291 (0.084)
All workers' earnings pretrend, 1970–1980		0.008 (0.064)	−0.016 (0.060)		0.001 (0.109)	−0.035 (0.102)
State fixed effects (26)	✓	✓	✓	✓	✓	✓
R^2	0.676	0.654	0.676	0.690	0.667	0.690

Notes: Positive (negative) coefficient estimates for the regional tariff reduction imply larger increases (declines) in informal earnings or employment in regions facing larger tariff reductions. Outcomes calculated using Census data. There are 405 microregion observations. Regional earnings premia calculated controlling for age, sex, education, and industry of employment. Efficiency weighted by the inverse of the squared standard error of the dependent variable estimate. Pretrends computed for 1980–1991 and 1970–1980. Standard errors (in parentheses) adjusted for 90 mesoregion clusters.

(b) Table 4: informal employment and earnings

6.9 Table 5: Changing Worker Composition

Read:

- Table 5 tests whether changing worker composition explains the growing earnings effect.
- Panel A repeats the main earnings coefficients for 1995, 2000, 2005, and 2010.
- Panel B controls for individual fixed effects with fixed returns to worker characteristics.
- Panel C allows time-varying returns to worker characteristics.
- The coefficients still become more negative over time in both fixed-effect approaches.

Economic message: The growing earnings effect is not just mechanical selection of different workers into formal employment. Even after accounting for worker fixed effects, exposed regions experience relative earnings declines.

6.10 Figure 5: Regional Imports, Exports, and Net Exports

Read:

- Figure 5 shows how regional tariff reductions predict import, export, and net export changes per worker.
- Imports rise early in exposed regions, as expected.
- The import effect does not grow in the same way as the earnings effect from 1995 to the early 2000s.
- Exports also rise in exposed regions, which works against a simple story that deteriorating trade quantities drive earnings declines.

Economic message: Slow adjustment in import and export quantities does not match the timing of the growing earnings decline. The earnings profile must be driven by another mechanism.

Change in log formal earnings premia:	1991–1995 (1)	1991–2000 (2)	1991–2005 (3)	1991–2010 (4)
<i>Panel A. Main specification</i>				
Regional tariff reduction (RTR_t)	−0.096 (0.120)	−0.529 (0.141)	−1.294 (0.139)	−1.594 (0.169)
<i>Panel B. Earnings premia controlling for individual fixed effects (fixed returns)</i>				
Regional tariff reduction (RTR_t)	−0.193 (0.115)	−0.514 (0.144)	−1.119 (0.147)	−1.271 (0.172)
<i>Panel C. Earnings premia controlling for individual fixed effects (time-varying returns)</i>				
Regional tariff reduction (RTR_t)	−0.230 (0.093)	−0.551 (0.098)	−1.322 (0.094)	−1.454 (0.119)
Formal earnings pretrend, 1986–1990				
State fixed effects (26)	✓	✓	✓	✓

Notes: Negative coefficient estimates for the regional tariff reduction (RTR_t) imply larger declines in formal earnings in regions facing larger tariff reductions. Microregion observations: panel A, 475; panels B and C, 450 (omits regions with insufficient observations to identify region-year fixed effects in any particular year). Regional earnings premia: panel A: calculated controlling for age, sex, education, and industry of employment; panels B and C: controlling for individual fixed effects. Standard errors (in parentheses) adjusted for 112 mesoregion clusters. Efficiency weighted by the inverse of the squared standard error of the estimated change in log formal earnings premium. See text for detailed description of each panel.

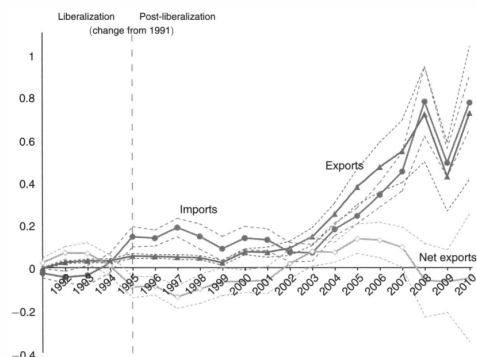


FIGURE 5. REGIONAL IMPORTS, EXPORTS, AND NET EXPORTS PER WORKER, 1991–2010

Notes: Each point reflects an individual regression coefficient, $\hat{\theta}_t$, following (3), where the dependent variable is the change in regional imports per workers (circles), exports per worker (triangles), or net exports per worker (diamonds), measured in \$100,000 units. The independent variable is the regional tariff reduction (RTR_t), defined in (2). Note that RTR_t always reflects tariff reductions from 1990 to 1995. All regressions include state fixed effects, but do not include pre-liberalization trends due to a lack of Comtrade trade data before 1989. Positive estimates imply larger increases in trade flow per worker in regions facing larger tariff reductions. Vertical bar indicate that liberalization was complete by 1995. Dashed lines show 95 percent confidence intervals. Standard errors adjusted for 112 mesoregion clusters.

(a) Table 5: worker composition

(b) Figure 5: trade quantities

6.11 Table 6: Slow Response of Imports or Exports

Read:

- Table 6 adds controls for regional import and export growth to the main earnings regressions.
- Panel B shows OLS controls for trade quantities; the RTR_t coefficients still grow from -0.089 in 1995 to -1.562 in 2010.
- Panels C and D instrument regional trade quantities using other countries' trade flows.
- The 2010 RTR_t coefficients remain large under both Latin America and Colombia IV strategies.

Economic message: Even when the paper controls for realized trade-flow changes, the dynamic earnings effect remains. Slow trade quantity adjustment is not the mechanism.

6.12 Figure 6: Inferred Adjustment and Capital Adjustment

Read:

- Figure 6 compares the inferred labor demand adjustment with proxies for capital adjustment.
- The inferred labor demand decline grows steadily in exposed regions.
- The capital-adjustment profiles also grow over time and have a similar scale.
- Depending on the fixed-factor share, capital adjustment can explain a large part of the 2010 inferred adjustment.

Economic message: The figure provides direct support for dynamic labor demand. Capital does not instantly adjust after trade liberalization; its gradual withdrawal from exposed regions deepens local earnings losses.

TABLE 6—MECHANISMS: SLOW RESPONSE OF IMPORTS OR EXPORTS, 1995, 2000, 2005, AND 2010

Change in log formal earnings premia:	1991–1995 (1)	1991–2000 (2)	1991–2005 (3)	1991–2010 (4)
<i>Panel A. Main specification</i>				
Regional tariff reduction (<i>RTR</i>)	−0.096 (0.120)	−0.529 (0.141)	−1.294 (0.139)	−1.594 (0.169)
<i>Panel B. Controls for trade quantities (OLS)</i>				
Regional tariff reduction (<i>RTR</i>)	−0.089 (0.112)	−0.521 (0.138)	−1.287 (0.181)	−1.562 (0.221)
Import quantity control			−0.382 (2.242)	
Export quantity control			0.142 (3.355)	
<i>Panel C. Latin America IV</i>				
Regional tariff reduction (<i>RTR</i>)	−0.129 (0.106)	−0.569 (0.129)	−1.342 (0.173)	−1.757 (0.212)
Import quantity control			1.668 (2.631)	
Export quantity control			−0.149 (3.861)	
First-stage <i>F</i> (Kleibergen-Paap)			93.04	
<i>Panel D. Colombia IV</i>				
Regional tariff reduction (<i>RTR</i>)	−0.049 (0.108)	−0.488 (0.132)	−1.372 (0.161)	−1.502 (0.213)
Import quantity control			−3.489 (2.427)	
Export quantity control			5.379 (3.268)	
First-stage <i>F</i> (Kleibergen-Paap)			876.2	
Formal earnings pretrend, 1986–1990	✓	✓	✓	✓
State fixed effects (26)	✓	✓	✓	✓

Notes: Negative coefficient estimates for the regional tariff reduction (*RTR*) imply larger declines in formal earnings in regions facing larger tariff reductions. Panel A replicates the earnings results in columns 3 and 6 of Table 2. Panels B–D include regional import and export quantity controls as in (7). We instrument for the potentially endogenous import and export controls using regional measures of commodity price growth from Adão (2016) and with regional trade flows for other countries. “Latin America” consists of Argentina, Chile, Colombia, Paraguay, Peru, and Uruguay. We measure imports and exports between Latin America or Colombia and the rest of the world excluding Brazil. Due to Comtrade data availability, changes in Colombian trade flows are measured from 1991 to each subsequent year and Latin American trade flows from 1994. We allow for time-varying first-stage coefficients, so we have 2 endogenous variables (*RegImp_{it}* and *RegExp_{it}*) and 57 instruments for Colombia (3 instruments $\times$ 19 years) and 48 instruments for Latin America (3 instruments $\times$ 16 years). First-stage Kleibergen-Paap *F*-statistics

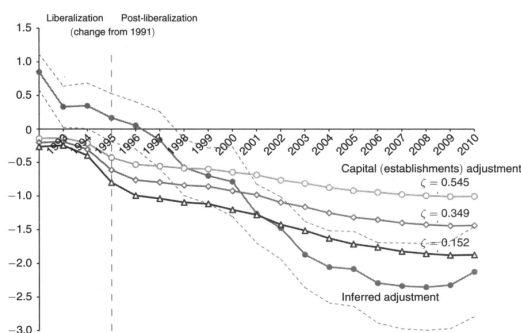


FIGURE 6. INFERRED ADJUSTMENT AND CAPITAL ADJUSTMENT QUANTIFICATION, 1992–2010

Notes: Each point reflects an individual regression coefficient, $\hat{\theta}_i$, following (3). For the profile with solid circles, the dependent variable is the inferred labor demand shifts from agglomeration and capital adjustment, defined in (10). For the gray profiles with hollow markers, the dependent variable is capital's contribution to overall adjustment, using the change in the number of regional formal establishments as a proxy for the change in regional capital, $\sum_i \lambda_{it} \bar{K}_{it}$. We present profiles for three values of ζ , specific factors' share of nonlabor inputs, based on Valentinyi and Herrendorf (2008). The independent variable is the regional tariff reduction (*RTR_{it}*), defined in (2). Note that *RTR_{it}* always reflects tariff reductions from 1990 to 1995. All regressions include state fixed effects, and post-liberalization regressions control for the 1986–1990 outcome pretrend. Negative estimates imply larger declines in residual labor demand or the number of establishments in regions facing larger tariff reductions. Vertical bar indicates that liberalization was complete by 1995. Dashed lines show 95 percent confidence intervals. Confidence intervals for capital adjustment profiles shown in online Appendix B.12. Standard errors adjusted for 112 mesoregion clusters.

(a) Table 6: trade quantity controls

(b) Figure 6: inferred and capital adjustment

6.13 Table 7: Test for Agglomeration Economies

Read:

- Table 7 estimates region-by-industry employment changes from 1991 to 2010.
- It controls for an industry's own tariff reduction and then asks whether the rest of the region's tariff exposure matters.
- The coefficient on regional tariff reduction is negative and highly significant in every specification.

- This holds for all industries and for tradable industries only.

Economic message: An industry does worse in regions where other industries are more adversely hit. Without agglomeration, local workers leaving other industries should help the reference industry. The opposite pattern supports agglomeration-based local spillovers.

6.14 Figure 7: Formal Establishments and Average Establishment Size

Read:

- Figure 7 shows that regions facing larger tariff cuts experience steady relative declines in the number of formal establishments.
- Average formal establishment size also falls rather than rising.
- The establishment effect grows gradually and levels off later in the sample.
- The pre-liberalization patterns do not explain the post-liberalization decline.

Economic message: The number of establishments is a proxy for regional capital. The figure suggests that capital gradually moves away from exposed regions, rather than simply reallocating into larger surviving establishments.

	Change in log region $\times$ industry employment					
	All industries				Tradable industries	
	(1)	(2)	(3)	(4)	(5)	(6)
Regional tariff reduction (RTR_t)	-7.751 (0.625)	-6.084 (0.623)	-6.183 (0.631)	-6.333 (0.646)	-6.708 (0.675)	-6.704 (0.694)
Industry tariff reduction	-1.790 (0.294)	-1.666 (0.290)	-1.669 (0.291)		-2.017 (0.332)	
Formal employment pretrend, 1986–1990			-0.106 (0.036)	-0.147 (0.032)	-0.110 (0.037)	-0.150 (0.032)
Industry fixed effects (20)		✓	✓	✓	✓	✓
State fixed effects (26)					✓	✓

Notes: Negative coefficient estimates for the regional tariff reduction imply the presence of agglomeration economies, following (15). Observations represent region $\times$ industry pairs. The dependent variable is the change in log formal employment in a given region $\times$ industry pair from 1991 to 2010. Columns 1–4 cover all industries, including the nontradable sector, while columns 5 and 6 restrict attention to tradable industries. For tradable industries, industry tariff reductions are given by the decline in the log of 1 plus the tariff rate. For the nontradable sector, the industry tariff reduction is measured using RTR_t . Standard errors (in parentheses) adjusted for 112 mesoregion clusters.

tariff changes, following the arguments in (Kovak 2013). The results of estimating

(a) Table 7: agglomeration test

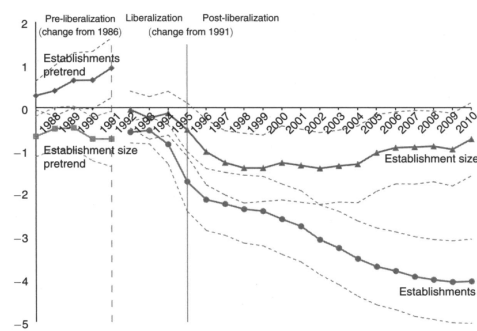


FIGURE 7. REGIONAL LOG NUMBER OF FORMAL ESTABLISHMENTS AND LOG AVERAGE FORMAL ESTABLISHMENT SIZE (NUMBER OF WORKERS), 1987–2010

Notes: Each point reflects an individual regression coefficient, $\hat{\theta}_t$, following (3), where the dependent variable is the change in regional log number of formal establishments or the change in regional log average formal establishment size. The independent variable is the regional tariff reduction (RTR_t), defined in (2). Note that RTR_t always reflects tariff reductions from 1990 to 1995. For circles and triangles, the changes are from 1991 to the year listed on the x-axis. For diamonds and squares, the changes are from 1986 to the year listed. All regressions include state fixed effects, and post-liberalization regressions control for the 1986–1990 outcome pretrend. Negative estimates imply larger declines in the number of establishments or average establishment size in regions facing larger tariff reductions. Vertical bars indicate that liberalization began in 1991 and was complete by 1995. Dashed lines show 95% confidence intervals.

(b) Figure 7: establishments and size

6.15 Figure 8: Formal Establishment Entry and Exit

Read:

- Figure 8 decomposes establishment adjustment into cumulative entry and exit.
- Establishment entry falls quickly in exposed regions after liberalization.
- Establishment exit rises gradually over time.
- This timing fits the idea that new investment can redirect quickly, while installed capital exits only slowly as depreciation and closure occur.

Economic message: The adjustment is asymmetric. New capital formation responds quickly, but old capital leaves slowly, generating a prolonged decline in exposed regions.

6.16 Figure 9: Job Creation and Destruction

Read:

- Figure 9 gives an analogous decomposition for jobs.
- Job creation falls quickly in regions with larger tariff reductions.
- Job destruction rises gradually and remains high.
- The pattern mirrors the establishment entry/exit evidence.

Economic message: The labor-market counterpart to capital adjustment is lower job creation and higher job destruction. This reinforces the view that exposed regions experience a slow deterioration of formal economic activity.

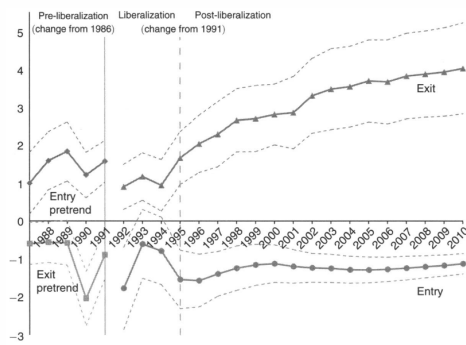


FIGURE 8. REGIONAL LOG CUMULATIVE FORMAL ESTABLISHMENT ENTRY AND EXIT, 1987–2010

Notes: Each point reflects an individual regression coefficient, $\hat{\theta}_t$, following (3). The dependent variable is the log cumulative formal establishment entry or exit from 1991 to the year listed on the x-axis (circles and triangles) or from 1986 to the year listed (diamonds and squares), calculated as in Davis and Haltiwanger (1990). The independent variable is the regional tariff reduction (RTR_t), defined in (2). Note that RTR_t always reflects tariff reductions from 1990 to 1995. All regressions include state fixed effects, and post-liberalization regressions control for log cumulative establishment entry or exit during the period 1986–1990. Positive exit estimates and negative entry estimates imply larger rates of exit and smaller rates of entry in regions facing larger tariff reductions. Vertical bars indicate that liberalization began in 1991 and was complete by 1995. Dashed lines show 95 percent confidence intervals. Standard errors adjusted for 119 sequential clusters.

(a) Figure 8: establishment entry and exit

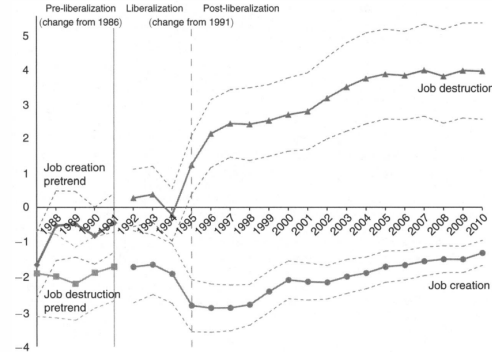


FIGURE 9. REGIONAL LOG CUMULATIVE JOB CREATION AND DESTRUCTION, 1987–2010

Notes: Each point reflects an individual regression coefficient, $\hat{\theta}_t$, following (3). The dependent variable is the log cumulative job creation or destruction rate from 1991 to the year listed on the x-axis (circles and triangles) or from 1986 to the year listed (diamonds and squares), calculated as in Davis and Haltiwanger (1990). The independent variable is the regional tariff reduction (RTR_t), defined in (2). Note that RTR_t always reflects tariff reductions from 1990 to 1995. All regressions include state fixed effects, and post-liberalization regressions control for log cumulative job creation or destruction during the period 1986–1990. Positive job destruction estimates and negative job creation estimates imply larger rates of job destruction and smaller rates of job creation in regions facing larger tariff reductions. Vertical bars indicate that liberalization began in 1991 and was complete by 1995. Dashed lines show 95 percent confidence intervals. Standard errors adjusted for 119 sequential clusters.

(b) Figure 9: job creation and destruction

6.17 Table 8: Agglomeration Elasticity Estimates

Read:

- Panel A estimates an inverse labor supply elasticity of 0.363 using RTR_r as an instrument for regional earnings changes.
- Panel B reports agglomeration elasticities for three assumptions about fixed factors' share of nonlabor inputs.
- Wage-based estimates range from 0.042 to 0.333.
- Employment-based estimates range from 0.215 to 0.461.
- The intermediate wage-based estimate of 0.188 is close to estimates in prior agglomeration literature.

Economic message: Table 8 shows that standard-sized agglomeration economies can help quantitatively explain the long-run regional earnings effects. The proposed mechanism is not just qualitatively plausible; it is quantitatively large enough.

TABLE 8—AGGLOMERATION ELASTICITY ESTIMATES			
<i>Panel A. Inverse labor supply elasticity (η)</i>	0.363 (0.060)		
<i>Panel B. Agglomeration elasticity (κ)</i>	low ζ (0.152) (1)	mid ζ (0.349) (2)	high ζ (0.545) (3)
Wage-based agglomeration elasticity	0.042 (0.023)	0.188 (0.023)	0.333 (0.025)
Employment-based agglomeration elasticity	0.215 (0.032)	0.330 (0.038)	0.461 (0.043)

Notes: Labor supply elasticity, η , estimated from (12) using *RTR*, an instrument for the change in regional log earnings premium. The first-stage partial *F*-statistic (Kleibergen-Paap) for this regression is 59.14. Given the estimate of η , the agglomeration elasticity, κ , is estimated using two alternative methods. The earnings-based approach estimates (13), and the employment-based approach estimates (14), both using nonlinear least squares, and both including 1986–1990 pre-liberalization outcome trends and state fixed effects. The employment-based estimates control for industry price changes as in column 3 of Table 7, and results using other approaches are very similar. We present estimates for three different values of ζ , specific factors’ share of nonlabor inputs, based on Valentini and Herrendorf (2008). See text for details. Standard errors (in parentheses) bootstrapped by regional resampling.

Table 8: agglomeration elasticity estimates

7 Short summary

- The paper studies Brazil’s early-1990s trade liberalization using annual administrative formal-labor-market data from 1986 to 2010.
- Regions are exposed to the shock according to their initial industry mix and the size of tariff cuts in those industries.
- Regions facing larger tariff cuts experience persistent declines in formal earnings and formal employment relative to other regions.
- The formal earnings effect grows for more than a decade and is more than three times larger after 20 years than after 10 years.
- Working-age population does not respond much, suggesting limited interregional migration.
- Informal employment rises in exposed regions, indicating that many workers leave formal jobs but remain in the same region.
- Worker composition, slow import/export responses, and post-liberalization shocks do not explain the growing earnings effect.
- Establishment counts, entry, exit, job creation, and job destruction show gradual deterioration in exposed regions.
- The supported mechanism is imperfect interregional labor mobility plus dynamic labor demand from slow capital adjustment and agglomeration economies.
- The key lesson is that short-run local estimates of trade shocks can understate long-run regional effects.

8 Conclusion

8.1 Core conclusion

Dix-Carneiro and Kovak show that the regional labor-market effects of trade liberalization can grow for many years after the policy reform. In Brazil, regions more exposed to tariff reductions experienced long-lasting relative declines in formal earnings and formal employment. These effects do not fade away through migration or standard spatial arbitrage. Instead, they become larger for more than a decade and level off only in the late 2000s.

8.2 Economic intuition

The key intuition is that workers and capital do not reallocate instantly across space. Workers in harder-hit regions often remain in the same region but move out of formal employment. Meanwhile, capital and establishments gradually move away or disappear through lower entry, lower job creation, higher exit, and higher job destruction. As formal activity declines, local productivity may also decline through weaker agglomeration forces. These mechanisms amplify the initial tariff shock over time.

8.3 Incremental contribution revisited

The paper advances the trade-and-labor literature by turning a static regional exposure design into a dynamic analysis. It shows that the timing of effects is itself an important empirical object. A medium-run estimate is not necessarily an upper bound on long-run costs; in this setting, it is a lower bound. The paper also connects reduced-form regional evidence to a mechanism involving capital adjustment and agglomeration.

8.4 Implications for trade policy

The results imply that the local costs of trade liberalization can be geographically persistent. Policy evaluation should account for long adjustment horizons, regional formal-informal transitions, and firm dynamics. Trade adjustment policies focused only on short-run displacement may miss the prolonged regional deterioration documented here.

8.5 Limitations

The analysis focuses primarily on the formal labor market because RAIS does not observe informal workers or nonemployment directly. Census data help address this limitation, but only at decennial frequency. The mechanism analysis uses proxies for regional capital, especially the number of formal establishments, because direct regional capital stock data are not available. The evidence is strong for Brazil's liberalization episode, but external validity depends on the degree of labor mobility, informality, capital adjustment, and agglomeration in other countries.

8.6 Takeaway

The enduring takeaway is that regional adjustment to trade can be slow, cumulative, and amplifying. The paper changes the interpretation of local trade-shock estimates: what looks like a medium-run effect may be only the early stage of a much larger long-run regional transformation.